

HACHINOHE AP, Japan

WMO: 475150

Lat: 40.552N

Lon: 141.467E

Elev: 49

StdP: 100.73

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-6.1	-12.0	1.3	-4.8	-11.0	1.5	-3.9	12.4	1.2	11.0	0.8	4.2	220	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	31.1	24.6	29.2	23.5	27.8	22.7	25.3	29.8	24.5	28.3	23.6	26.7	5.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.0	27.7	23.1	18.0	26.9	22.2	17.0	26.0	77.7	29.8	74.4	28.2	70.9	26.3	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.7	8.6	DB	-9.4	34.1	1.3	1.5	-10.3	35.2	-11.0	36.1	-11.7	36.9	-12.7	38.0	(4)
(5)			WB	-10.0	26.5	1.1	0.9	-10.8	27.1	-11.5	27.6	-12.2	28.1	-13.0	28.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-0.9	-0.2	3.1	8.9	14.0	17.1	21.4	23.1	19.8	13.6	7.2	1.6	(6)	
	DBStd	8.87	2.54	2.94	3.33	3.67	3.60	3.16	3.60	3.15	3.14	3.06	3.63	2.92	(7)	
	HDD10.0	1268	336	284	214	64	7	0	0	0	0	5	97	259	(8)	
	HDD18.3	3098	594	518	471	284	140	60	12	2	18	149	333	517	(9)	
	CDD10.0	1561	0	0	2	30	132	212	354	406	294	117	14	1	(10)	
	CDD18.3	349	0	0	0	1	6	22	107	149	62	2	0	0	(11)	
	CDH23.3	2069	0	0	0	7	47	101	646	963	301	4	0	0	(12)	
	CDH26.7	564	0	0	0	1	7	18	184	287	67	0	0	0	(13)	
(14) Wind	WSAvg	4.1	4.4	4.5	4.8	4.8	4.2	3.5	3.4	3.5	3.6	3.9	4.2	4.3	(14)	
(15) Precipitation	PrecAvg	1173	67	54	67	70	90	102	142	143	165	110	85	72	(15)	
	PrecMax	1483	131	153	116	148	142	203	277	328	322	217	163	189	(16)	
	PrecMin	886	27	20	21	15	32	32	24	26	54	27	28	36	(17)	
	PrecStd	159	27	28	27	33	34	41	64	83	67	55	37	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	11.7	17.0	23.9	27.2	29.1	33.2	34.0	31.2	23.9	19.1	12.9	(19)	
		MCWB	5.0	7.4	9.9	14.1	16.8	20.4	25.2	25.9	24.8	18.1	14.2	10.1	(20)	
	2%	DB	5.9	8.1	13.8	20.0	24.2	26.0	30.9	31.9	28.5	21.9	16.8	10.1	(21)	
		MCWB	3.0	4.6	8.4	12.3	16.0	19.5	24.4	25.1	23.5	16.8	12.7	7.4	(22)	
	5%	DB	4.2	6.2	11.0	17.2	22.1	24.0	28.9	29.9	26.2	20.2	15.0	8.1	(23)	
		MCWB	1.7	2.9	6.7	10.4	15.0	18.5	23.4	24.3	21.6	15.9	11.5	5.6	(24)	
	10%	DB	3.1	4.8	9.0	15.1	20.1	22.1	26.9	28.0	24.9	19.0	13.2	6.2	(25)	
		MCWB	0.8	1.9	5.3	9.7	14.3	17.8	22.5	23.4	21.0	15.1	10.1	3.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	7.9	10.9	15.2	18.6	21.9	25.9	26.5	25.7	19.7	15.2	10.7	(27)	
		MCDB	7.1	10.4	15.7	21.1	24.2	25.5	31.6	31.7	29.7	21.8	17.5	12.4	(28)	
	2%	WB	3.3	5.0	8.7	13.2	17.3	20.5	24.7	25.7	24.3	17.9	13.3	8.0	(29)	
		MCDB	5.1	7.6	12.7	17.9	22.1	24.5	29.4	30.3	27.4	20.8	16.1	9.8	(30)	
	5%	WB	2.1	3.3	7.1	11.6	16.0	19.4	23.9	24.8	22.9	16.8	11.9	5.7	(31)	
		MCDB	3.7	5.8	10.3	16.1	20.8	23.2	27.8	28.8	25.6	19.3	14.4	7.7	(32)	
	10%	WB	1.0	1.9	5.4	10.3	14.9	18.3	23.0	24.1	21.5	15.7	10.6	4.1	(33)	
		MCDB	2.6	4.2	8.4	14.2	19.3	21.7	26.2	27.2	24.0	18.3	12.8	6.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	7.2	8.1	9.4	8.8	7.2	6.6	6.6	7.7	9.0	8.3	6.6	(35)	
		MCDBR	7.7	9.7	11.5	12.6	12.4	10.3	9.6	9.4	9.3	10.1	10.8	9.0	(36)	
	5% WB	MCWBR	5.9	7.1	7.8	7.2	6.3	5.2	4.5	4.3	4.9	6.2	7.6	7.1	(37)	
		MCWBR	7.5	9.4	11.0	11.2	11.0	9.1	8.9	8.4	8.1	8.4	9.8	8.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.365	0.408	0.479	0.527	0.542	0.524	0.525	0.498	0.442	0.415	0.395	0.375	(40)	
		taud	2.062	1.940	1.855	1.828	1.869	1.997	2.066	2.153	2.265	2.214	2.161	2.105	(41)	
	Ebn at Noon		751	771	759	755	755	770	764	774	795	767	717	704	(42)	
		Edh at Noon	115	151	184	203	199	175	163	145	122	114	105	103	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.59	2.30	3.38	4.45	4.92	4.82	4.29	4.12	3.57	2.79	1.85	1.36	(44)	
	RadStd		0.07	0.19	0.25	0.40	0.51	0.36	0.48	0.51	0.29	0.24	0.14	0.09	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page